

Handshaker Utility

Handshaker is a Python utility for calculating photometric corrections (magcor) associated with Roman WFI calibration effects and for testing whether polynomial approximations to the wavelength-dependent ice throughputs are sufficiently accurate. The utility should be importable as a Python module and executable as a standalone script. A configuration file will be copied into the output directory for reproducibility.

Inputs

Global Inputs

- Calibration star SED
- Flag
 - TRUE: use the true ice model.
 - POLYFIT: use the polynomial approximation to the ice model.
- Filter transmission source
 - Temporary/fake filter set for testing.
 - Eventually the official SOC filter transmissions.
- Reference filter transmission
 - Defines the reference throughput against which wavelength-dependent changes are calculated.
- Output directory
 - Location for all tables, logs, plots, and configuration files.
- SN SED / model
 - For testing the chromatic difference between the calibration-star SED and the SN SED.

Observation-Specific Inputs

- Band
- MJD
 - Used to determine the appropriate ice thickness/model.
- RA, DEC
 - Used to determine the position on the focal plane.
- SCA
 - This is a required variable but will be internally checked against the SCA implied by the observation position.
- Source SED
 - Restricted to the relevant filter passband. (Won't break anything if not restricted)

Filter Transmission Handling

Filter transmission should be handled separately from the main calculation.

A `Filter/FilterSource` class should provide a common interface: `get_filter(band, MJD, source)`. The source determines where the transmission comes from.

SOC → `get_filter_SOC()`

Thushara → `get_filter_thushara()`

The MJD-dependent portion of the filter lookup should remain encapsulated within this interface. Explicitly print/log the filters being used

Focal-Plane / SCA Handling

Given (RA, DEC), Handshaker should determine the corresponding (x, y) position on the Roman focal plane. The supplied SCA should then be checked against the SCA implied by the position. If the two disagree, the utility should issue an error/warning rather than silently proceeding. Abort here? How problematic is it if they disagree?

Note: This is new. Previous versions of handshaker have not operated at the pixel level.

Corrections

Filter wavelength shifts

- SCA-dependent wavelength shifts.
- Ice-dependent wavelength shifts.
- Potentially other wavelength-dependent throughput changes. CRNL?

Ice correction

The ice thickness is determined from the observation MJD. The calculation can then use either (depending on config flag):

TRUE → true ice model

POLYFIT → polynomial approximation

SED-dependent chromatic correction (like done in DES)

The correction should account for the difference between:

- the calibration-star SED used to characterize the filter response, and
- the SN SED being observed.

Output Modes

Handshaker should have two primary modes.

Mode 1: magcor

Calculate the photometric correction for each observation and each effect.

Output

For each observation, return:

- MAGCOR_
- MAGCOR_TOTAL

Mode 2: polyfit

Evaluate the polynomial approximation to the throughput ratio.

The output should contain:

- Wavelength correction (WAVECOR).
- Polynomial order.
- Polynomial coefficients.
- Throughput ratio.
- Polynomial prediction.
- Residual.

Conceptually this is: $T(\lambda) / T(\lambda_{\text{ref}})$ is fit with an Nth-order polynomial.

Output table

ROW	SCA	ICE_THICK	BAND	WAVE	RATIO_TRUE	RATIO_POLY	RESIDUAL
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This allows plots to be generated directly from the output table without rerunning the calculation.

Polynomial Approximation Validation

A major purpose of Handshaker is to determine whether the polynomial approximation is sufficiently accurate for the science analysis.

For a given observation, calculate both:

True model:

mag_true

Polynomial approximation:

mag_poly

and define: $\text{mag_err} = \text{mag_poly} - \text{mag_true}$

The error captures the impact of replacing the true wavelength-dependent model with the polynomial approximation, including the associated wavelength correction.